

CIS 371 Web Application Programming

Fetch, Axios & Web Services



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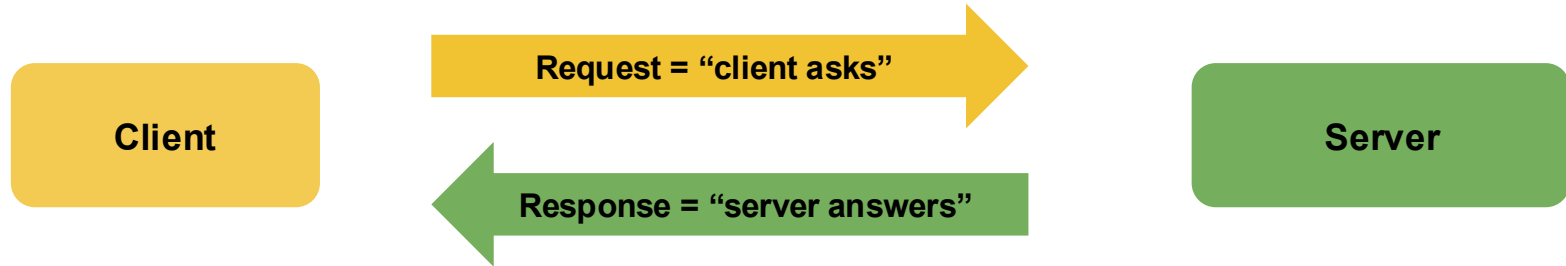
Topics

- HTTP requests and responses
- REST-style APIs and resources
- Sending GET requests with fetch and Axios
- Reading response data
- Sending POST requests and payloads
- Browser security: same-origin policy and CORS

What problem are we solving?

- A web app often needs data from another server
 - Example: weather data, user data, quotes, product info
- We use HTTP requests to ask for that data

HTTP Request & Response



HyperText: payload in response can be contents in **any** format (CSS, HTML, JPG, JS, JSON, PDF, PNG, ZIP,)

REST API

A **REST API** is a set of web URLs plus HTTP methods that let the frontend and backend exchange data in a standard way. It treats data as resources, and each resource is usually identified by a URL.

- A server exposes data/resources through URLs
- We use HTTP methods to tell the server what we want
 - GET → read data
 - POST → create new data
 - PUT / PATCH → update data
 - DELETE → remove

*REST is a design style, not a programming language or library. Usually, the data is sent back in **JSON** format.*

REST API

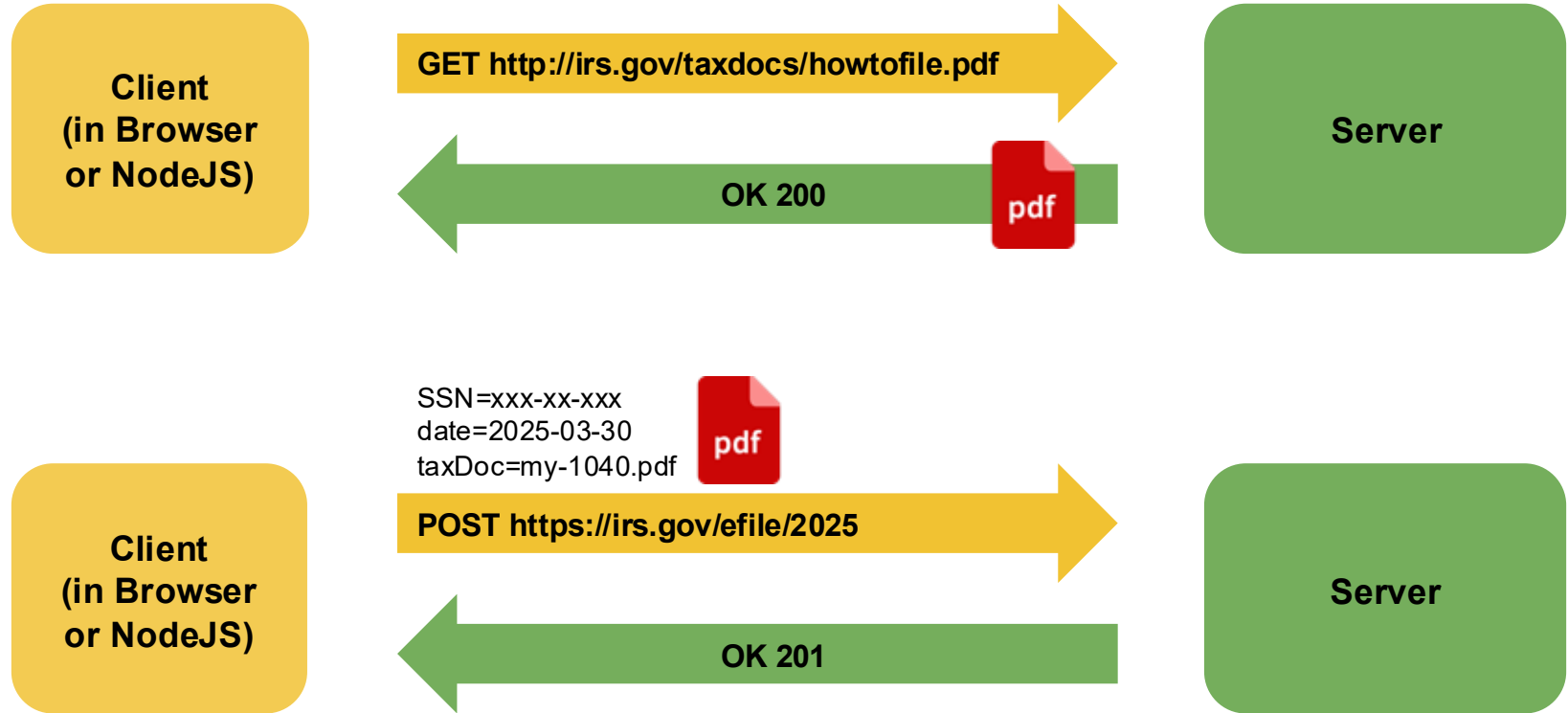
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- We use HTTP methods to tell the server what we want
 - GET → read data
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 - DELETE → remove

*REST is a design style, not a programming language or library. Usually the data is sent back in **JSON** format.*

GET /students → get all students
GET /students/5 → get student with id 5
POST /students → create a new student
DELETE /students/5 → delete student 5

GET and POST in Action



URL components

`https://blog.logrocket.com/wp-content/plugins/assets/rocket-logo.png`

Protocol *Host name* *Path* *resource name*

`https://px.linkedin.com/api/collect?pid=14682&fmt=png&allowSave=true`

query parameters

Query parameters are commonly used with GET requests.

```
{  
  pid: 14682,  
  fmt: "png",  
  allowSave: true  
};
```


Helpful Tools for Testing APIs



REST Client v0.25.1

Huachao Mao | 3,913,694 | ★★★★★ (348)

REST Client for Visual Studio Code

[Disable](#) [Uninstall](#)  

Extension is enabled on 'Container mcr.microsoft.com/devcont'



JSON Formatter

 callumlocke.com  **Featured**

Makes JSON easy to read. Open source.

★★★★★ 1,794 Developer Tools



Basic Pattern: Fetch vs Axios

```
fetch("http://some.website.org")
  .then((r: Response) => {
    //
    // Process the response
    // from the server here
    //
  })
  .catch((e: any) => {
    //
    // Handle potential
    // errors here
    //
  });
```

```
import axios, { AxiosResponse } from "axios";
axios
  .get("http://some.website.org")
  .then((r: AxiosResponse) => {
    //
    // Process the response
    // from the server here
    //
  })
  .catch((e: any) => {
    // Handle potential
    // errors here
  });
```

Both return a Promise

Which one?

Both are tools for sending HTTP requests to web services/APIs.

Axios (library, browser + Node.js)



Fetch (built into modern browsers; also available in modern Node.js, depending on version/environment)

node-fetch (npm lib)

GET with fetch vs Axios

```
axios.get(url).then(resp => resp.data)
```

```
fetch(url).then(r => r.json())
```

Axios gives parsed data in `resp.data`; fetch gives a `Response` object, so we parse separately.

Part A: Sending HTTP GET Requests

Send a GET Request

```
npm install axios
```

```
fetch("http://info.cern.ch")  
  .then((resp: Response) => {  
    console.log(resp.headers)  
    return resp.text()  
  })  
  .then((whatsInIt: string) => {  
    console.log(whatsInIt)  
  })
```

```
import axios, { AxiosResponse } from "axios";  
axios  
  .get("http://info.cern.ch")  
  .then((resp: AxiosResponse) => {  
    console.log(resp.headers);  
    return resp.data;  
  })  
  .then((whatsInIt: any) => {  
    console.log(whatsInIt);  
  });
```

Example Public APIs / Web Services

These are API providers we can use for practice.

- Random Users
 - Documentation: <https://randomuser.me/documentation>
 - Service Endpoint: <https://randomuser.me/api>
- Random Quotes
 - Documentation: <https://github.com/lukePeavey/quotable>
 - Service Endpoint: <https://api.quotables.io>
- A gazillion more Web Services: <https://github.com/public-apis/public-apis>
 - Pick ones that allow CORS

Example #1: Random User

Browser

<https://randomuser.me/api>

```
type RandUserData = {  
  results: Array<RandomUser>;  
  info: any;  
};  
type RandomUser = {  
  name: {  
    title: string;  
    first: string;  
    last: string;  
  };  
  email: string;  
};
```

Important: define these types to match the JSON structure of the API response.

```
axios  
  .request({  
    method: "GET",  
    url: "https://randomuser.me/api",  
  })  
  .then((resp: AxiosResponse) => resp.data)  
  .then((incoming: RandUserData) => {  
    console.log(incoming.info);  
    for (let k = 0; k < incoming.results.length; k++) {  
      console.log(incoming.results[k]);  
    }  
  });
```


Example #2: Random Quote

Browser

<http://api.quotable.io/random>

```
type Quote = {  
  tags: Array<string>;  
  content: string;  
  author: string;  
  length: number;  
};
```

```
axios  
  .request({  
    method: "GET",  
    url: "http://api.quotable.io/random",  
  })  
  .then((resp: AxiosResponse) => resp.data)  
  .then((q: Quote) => {  
    console.log(`${q.content} [${q.author}]`);  
  });
```

Example #3: Random User with Query Params

Browser

<https://randomuser.me/api?results=5&nat=gb,fr&inc=name,email,picture>

```
type RandUserData = {  
  results: Array<RandomUser>;  
  info: any;  
};  
type RandomUser = {  
  name: {  
    title: string;  
    first: string;  
    last: string;  
  };  
  email: string;  
};
```

```
axios  
  .request({  
    method: "GET",  
    url: "https://randomuser.me/api",  
    params: {  
      results: 5,  
      nat: "gb,fr",  
      inc: "name,email,picture",  
    },  
  })  
  .then((resp: AxiosResponse) => resp.data)  
  .then((incoming: RandUserData) => {  
    console.log(incoming.info);  
    for (let k = 0; k < incoming.results.length; k++) {  
      console.log(incoming.results[k]);  
    }  
  });
```

Example #4: Random Quotes with Query params

Browser

<http://api.quotable.io/quotes?limit=3>

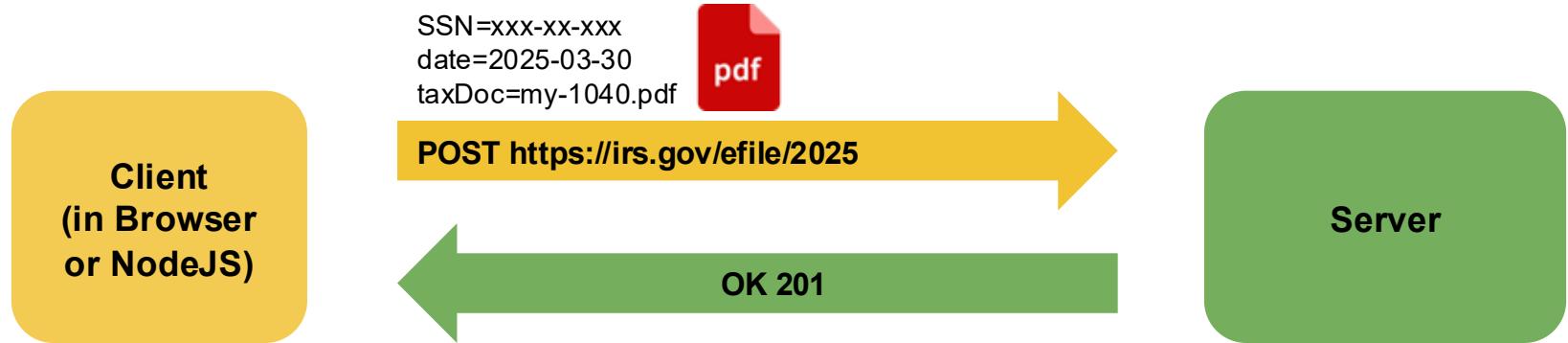
```
type QuoteResponse = {  
  count: number;  
  results: Array<Quote>;  
};  
type Quote = {  
  tags: Array<string>;  
  content: string;  
  author: string;  
  length: number;  
};
```

```
axios  
  .request({  
    method: "GET",  
    url: "http://api.quotable.io/quotes",  
    params: {  
      limit: 3,  
    },  
  })  
  .then((resp: AxiosResponse) => resp.data)  
  .then((qr: QuoteResponse) => {  
    console.log(qr.count);  
    for (let q of qr.results) {  
      console.log(q);  
    }  
  });
```



Part B: Sending HTTP POST Requests

HTTP Post Request



Example: HTTP POST Request + Data Payload

The screenshot shows the 'Headers' tab in Chrome DevTools. The status is '200 OK'. The request is a POST to 'https://aa.google.com/u/0/_/gog/get?rt=j&sourceid=538'. The 'Content-Length' header is highlighted with a yellow box and shows a value of 69.

POST https://aa.google.com/u/0/_/gog/get?rt=j&sourceid=538

Status: 200 OK

Version: HTTP/2

Transferred: 9.23 KB (11.17 KB size)

Referrer Policy: origin

Response Headers (766 B)

Request Headers (1.379 KB)

- Accept: */*
- Accept-Encoding: gzip, deflate, br
- Accept-Language: en-US,en;q=0.5
- Cache-Control: no-cache
- Connection: keep-alive
- Content-Length: 69**
- Content-Type: application/x-www-form-urlencoded;charset=utf-8

The screenshot shows the 'Request' tab in Chrome DevTools. It displays the 'Form data' and the 'Request payload'. The payload is a URL-encoded string: 'f.req=%5B%22og.botreq%22%2Cnull%2C%22%22%2Cnull%2Ctrue%2C0%2Cfalse%5D'. A yellow bracket indicates the payload size is 69 bytes.

Request payload: 69 bytes

1 f.req=%5B%22og.botreq%22%2Cnull%2C%22%22%2Cnull%2Ctrue%2C0%2Cfalse%5D

Important options for POST

```
const options = {
  method: "POST",
  url: "https://aa.google.com/u/0/_____",
  data: "discussed-in-the-next-few-slides",
  headers: {
    "Content-Type": "discussed-in-the-next-few-slides",
  },
};
axios.request(options).then((r: Response) => {
  /* your code here */
});
```

HTTP Post Data Payload

Content-Type	Format of Data Payload	When to Use
application/x-www-form-urlencoded	Key-value pair, special characters are encoded using ASCII Hex	Multiple textual data items of relatively small size
multipart/form-data	Multiple “documents”, delimited by special lines. Binary data are encoded to hex	Multiple text or binary data items of larger size (images, PDF,), each item becomes one attachment of your “email”
application/json	JSON (converted to text)	Structured textual data
text/plain	Any plain text	Unstructured textual data

- *The request should include “Content-Type” header to inform server how to parse/unpack the data payload*
- *JSON is the easiest to use*
- *Plain text is rarely used*

HTTP POST: plain-text payload

```
const myMessage: string = "Hello World\n" +  
  "Do you copy?";  
axios  
  .request({  
    method: "POST",  
    url: "https://pizza4.me/order",  
    headers: {  
      "Content-Type": "text/plain",  
    },  
    data: myMessage,  
  })  
  .then((r: Response) => {  
    /* code here */  
  });
```

Rarely used in practice

Actual HTTP Request

```
POST /order HTTP 1.1  
Host: https://pizza4.me  
Content-Type: text/plain  
Content-Length: 24  
Hello World  
Do you copy?
```


HTTP POST: JSON payload

```
const pizzaOrder = {
  size: 8,
  crust: "thin",
  toppings: ["cheese", "green pepper", "ham"],
  extraCheese: true,
};

axios
  .request({
    method: "POST",
    url: "https://pizza4.me/order",
    headers: {
      "Content-Type": "application/json",
    },
    data: JSON.stringify(pizzaOrder),
  })
  .then((r: Response) => {
    /* code here */
  });
```

- Use `JSON.stringify()` to convert a JS/TS object to its string representation
- The data is delivered as plain/text

Actual HTTP Request

```
POST /order HTTP/1.1
Host: https://pizza4.me
Content-Type: application/json
Content-Length: 89

{"size":8,"crust":"thin","toppings":
["cheese","green pepper","ham"],
"extraCheese":true}
```

HTTP POST: application/x-www-form-urlencoded

```
const payload = new URLSearchParams();
payload.append("size", "8"); // must be a string
payload.append("crust", "thin");
payload.append("extraCheese", "true");
const toppings = ["cheese", "green pepper", "ham"];
for (let k = 0; k < toppings.length; k++) {
  payload.append("toppings[]", toppings[k]);
}

axios
  .request({
    method: "POST",
    url: "https://pizza4.me/order",
    headers: {
      "Content-Type": "application/x-www-form-urlencoded",
    },
    data: payload,
  })
  .then((r: AxiosResponse) => {
    /* code here */
  });
```

- Key name for arrays ends with [] (empty brackets)

Actual HTTP Request

```
POST /order HTTP/1.1
Host: https://pizza4.me
Content-Type: application/x-www-form-urlencoded
Content-Length: 93

size=8&crust=thin&extraCheese=true&toppings[]=cheese&toppings[]=green%20pepper&toppings[]=ham
```



HTTP POST: multipart/form-data

```
import FormData from "form-data";

const payload = new FormData();
payload.append("size", 8);
payload.append("crust", "thin");
const toppings = ["cheese", "green pepper", "ham"];
toppings.forEach((t) => {
  payload.append("toppings[]", t);
});

axios
  .request({
    method: "POST",
    url: "https://pizza4.me/order",
    headers: payload.getHeaders(),
    data: payload,
  })
  .then((r: AxiosResponse) => {
    /* code here */
  });
```



HTTP POST: multipart/form-data

```
import FormData from "form-data";

const payload = new FormData();
payload.append("size", 8);
payload.append("crust", "thin");
const toppings = ["cheese", "green pepper", "ham"];
toppings.forEach((t) => {
  payload.append("toppings[]", t);
});

axios
  .request({
    method: "POST",
    url: "https://pizza4.me/order",
    headers: payload.getHeaders(),
    data: payload,
  })
  .then((r: AxiosResponse) => {
    /* code here */
  });
```

- Use this for sending multiple data which can be expressed as key-value pairs of significantly large size
- Each data item can be text or binary
- The HTTP protocol enforces a limit on the maximum message size. A huge payload must be split into smaller chunks

HTTP POST: multipart/form-data

Actual HTTP Request

```
POST /order HTTP/1.1
Host: https://pizza4.me
Content-Type: multipart/form-data; boundary=letseatpizza
Content-Length: xxxx

--letseatpizza
Content-Disposition: form-data; name="size"
Content-type: text/plain

8
--letseatpizza
Content-Disposition: form-data; name="toppings[]"

cheese
--letseatpizza
Content-Disposition: form-data; name="toppings[]"

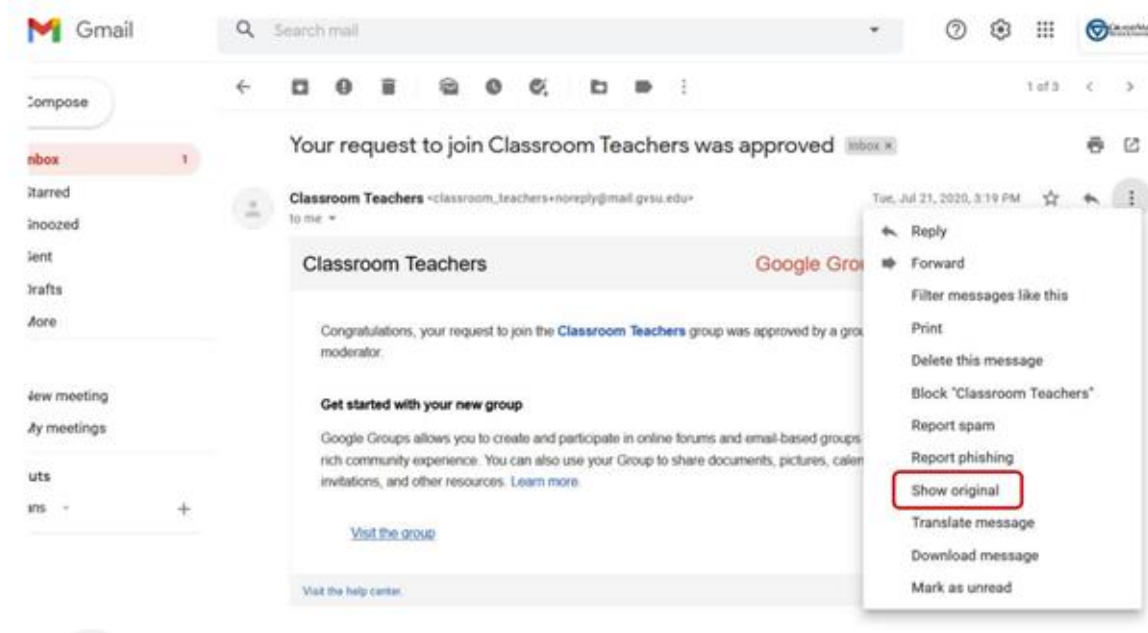
green pepper
--letseatpizza
Content-Disposition: form-data; name="extraCheese"

true
--letseatpizza--
```

- The boundary text (“letseatpizza” in example) will be auto generated by the utility you use, axios
- Each part may include more headers.
- Parts with binary data will include Content-Type header with proper value (such as “image/jpeg”) to inform the server how to unpack each part

HTTP POST: multipart/form-data

Try this: For a real example of a multipart message, open a message that has attachments in GMail and select “Show Original”

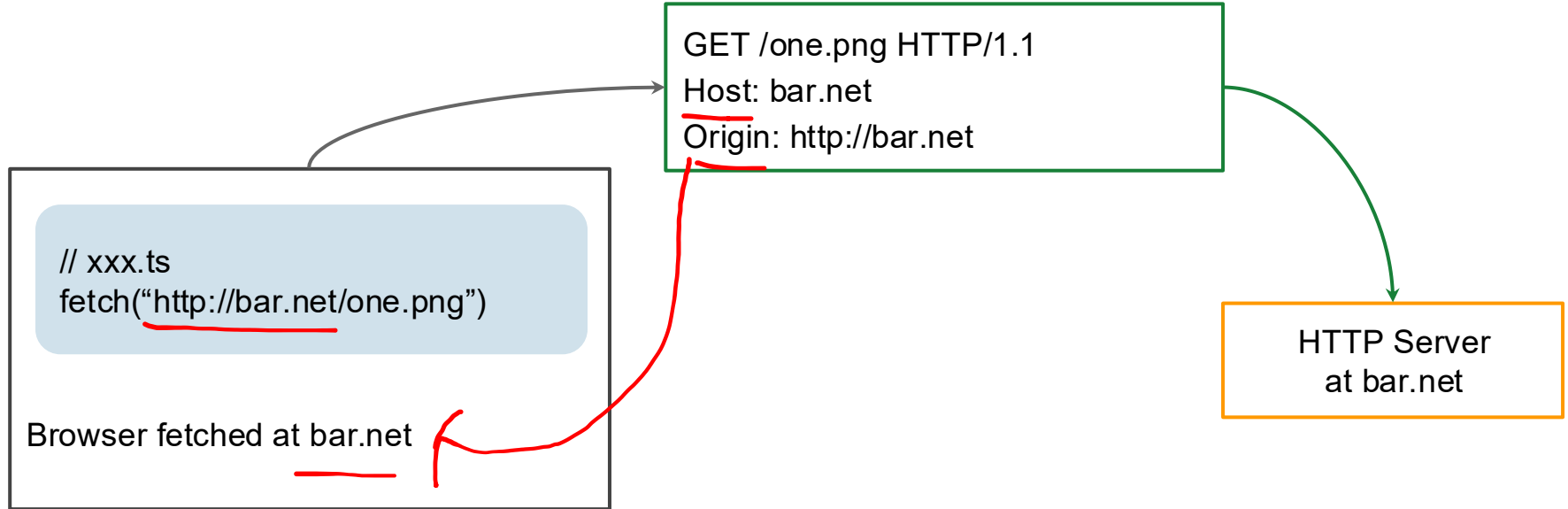


Security Issues

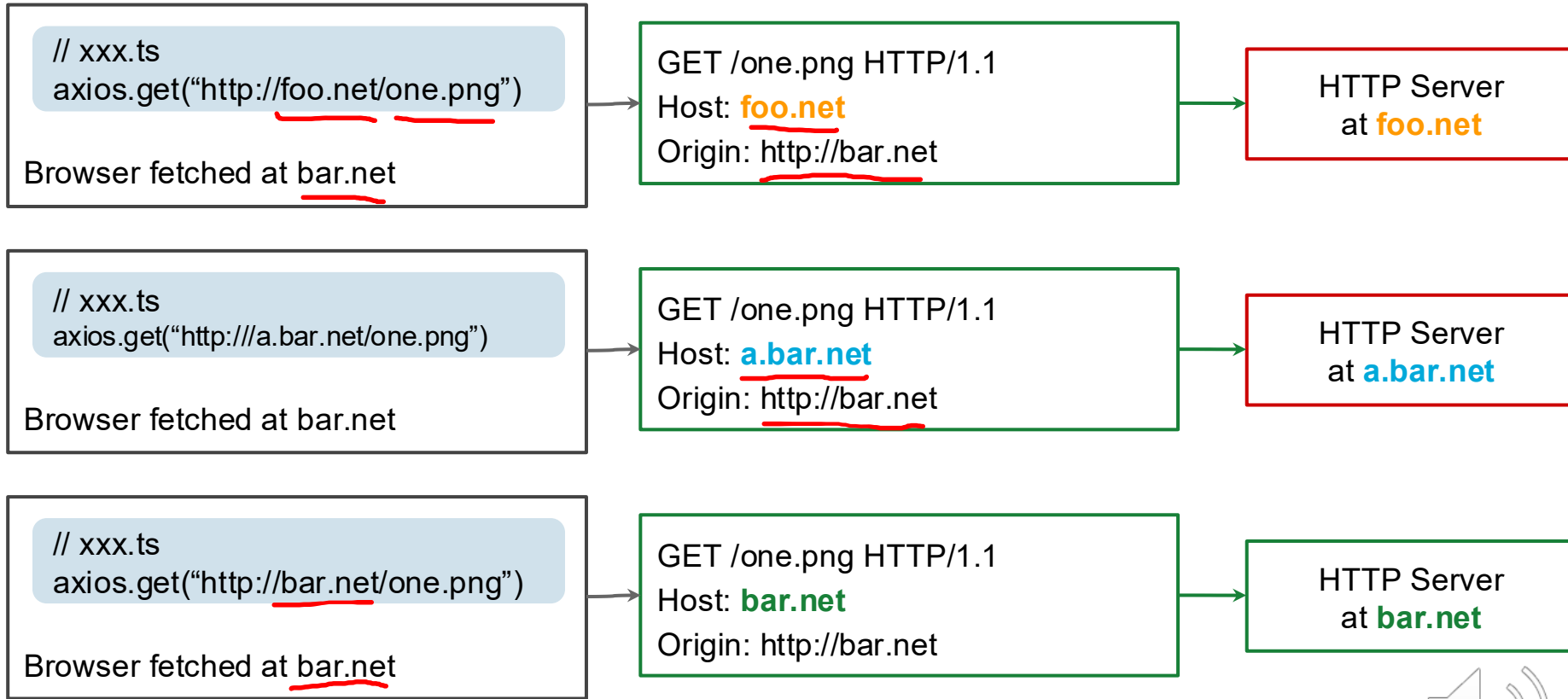
Browser Same-Origin Policy

- Origin = protocol + domain + port
- Scripts loaded from `http://some.domain.net` are allowed to fetch resources from `http://some.domain.net`
- Scripts loaded from **`http://some.domain.net`** are NOT automatically allowed to fetch from (cross-origin)
 - **`https://some.domain.net`** (different protocol)
 - `http://some.domain.org` (different domain)
 - `http://some.other-name.net` (different domain)
 - `http://some.domain.net:9000` (different port)
- Cross-origin requests require special handling by the server!

Same Origin



Different Origin (Cross Origin)

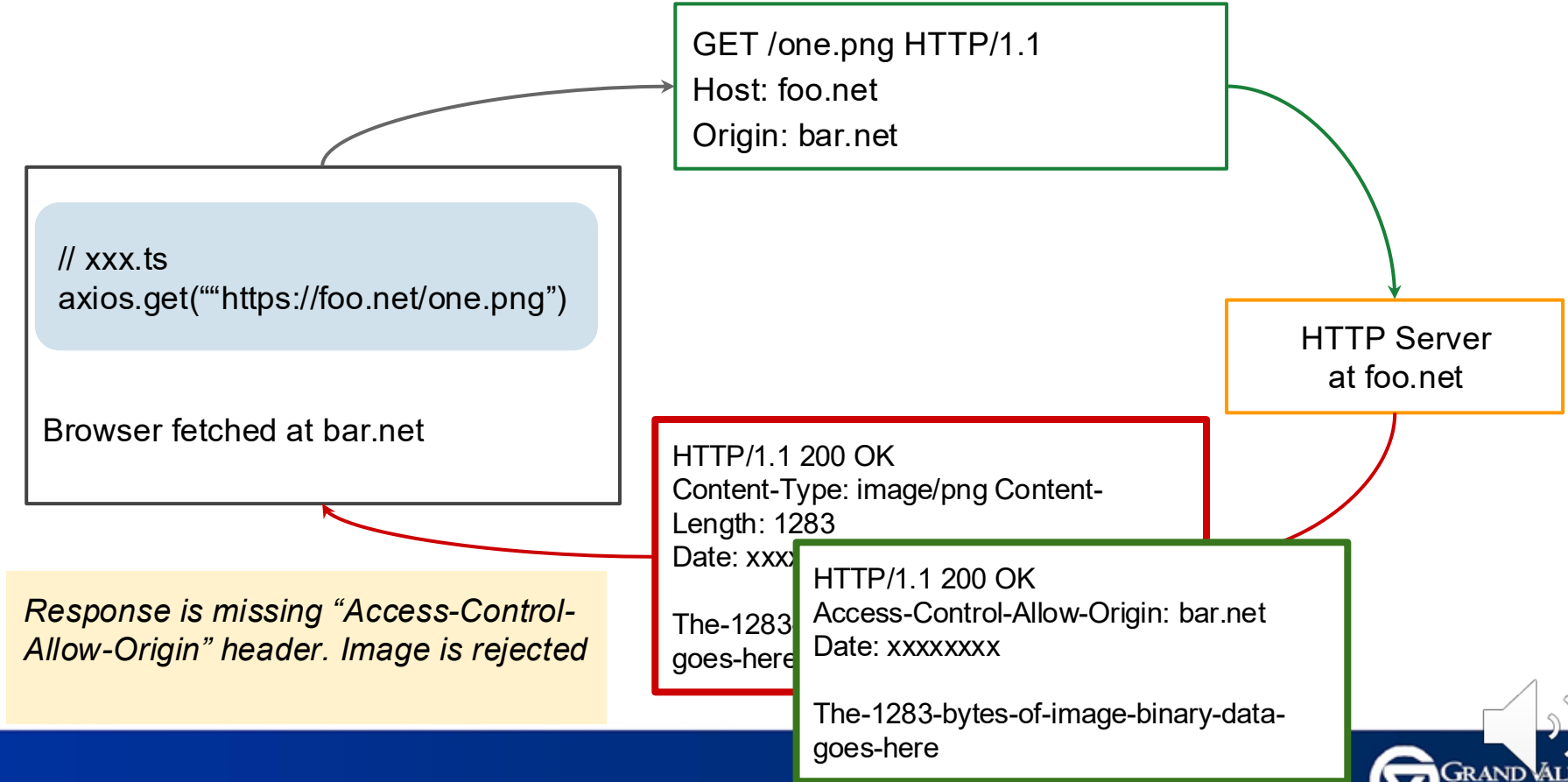


CORS: Controlled Cross-Origin Access

- CORS (Cross-Origin Resource Sharing):
- Same-origin policy blocks JavaScript from reading many cross-origin responses
- CORS allows servers to explicitly permit certain cross-origin requests
- Browser checks CORS response headers before exposing the response to JavaScript

What is CORS?

Simple CORS Request



Simple CORS Request



GET /one.png HTTP/1.1
Host: foo.net
Origin: bar.net

For what security reason, do we need same-origin policy

```
// xxx.ts  
axios.get("https://foo.net/one.png")
```

Browser fetched at bar.net

HTTP Server
at foo.net

HTTP/1.1 200 OK
Content-Type: image/png Content-
Length: 1283
Date: xxx

The-1283-
goes-here

HTTP/1.1 200 OK
Access-Control-Allow-Origin: bar.net
Date: xxxxxxxx

The-1283-bytes-of-image-binary-data-
goes-here

Response is missing "Access-Control-Allow-Origin" header. Image is rejected

Complex cross-origin requests

Cross-origin requests are complex if they use any of the following:

- Methods other than *GET*, *POST*, or *HEAD*
- Headers other than *Accept-Language*, *Accept*, or *Content-Language*
- Content-Type headers other than *multipart/form-data*, *application/x-www-form-urlencoded*, or *text/plain*

Preflight CORS Flow

Browser wants to send a cross-origin request

Browser first sends an **OPTIONS** request

Request may include:

- **Access-Control-Request-Method**

Server replies with:

- **Access-Control-Allow-Methods**

If allowed, browser sends the real request

GET https://randomuser.me/api?results=7&inc=name,email,picture

Status: HTTP/1.1 200

Request Headers

Accept	text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,image/apng,*/*;q=0.8
Accept-Encoding	gzip, deflate, br
Accept-Language	en-US,en;q=0.9,id-ID;q=0.8,id;q=0.7,la;q=0.6
Cookie	__cfduid=d57b18855239032023e9c7617b85ef10c1538271818
Referer	
Upgrade-Insecure-Requests	1
User-Agent	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_13_6) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/69.0.

Response Headers

access-control-allow-origin	*
cache-control	no-cache
cf-ray	463b7ca8bd607e75-DTW
content-encoding	br
content-type	application/json; charset=utf-8
date	Wed, 03 Oct 2018 01:03:05 GMT
etag	W/"878-sPQkdJ3SNfIA02yPG/UQdA"
expect-ct	max-age=604800, report-uri="https://report-uri.cloudflare.com/cdn-cgi/beacon/expect-ct"
server	cloudflare
status	200

Solving CORS issue

- Change the server settings to enable CORS (only possible if you are the admin of that server)
- Access the service via a middleware
 - Your script (in the browser) sends the request to a middleware (running on the same host where keep the script)
 - The middleware then sends the actual request to the actual server. This strategy tricks the Browser as if the responses are coming from the middleware running on the same host
- Using a third-party Proxy server (in place of your own middleware)